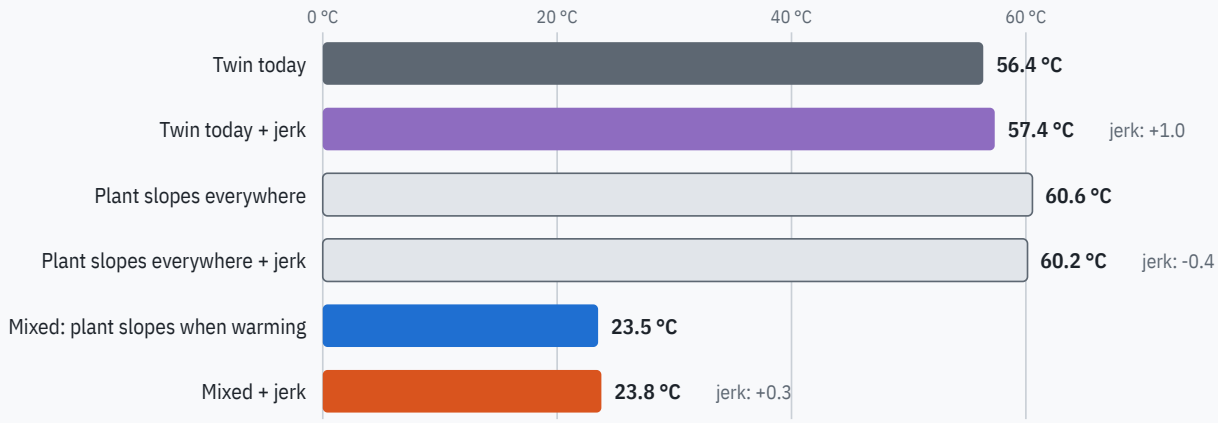
SOFC PLANT TWIN · JERK TEST AGAINST THE POC DATA

What if the twin gets the jerk?

Almost nothing changes. Adding the jerk moves the twin's average gap to the plant by about 1 °C, whichever version of the twin we add it to. The twin's problem is that it is missing heat, not that its corners are too sharp.

“Jerk” here means: at a stage border the stack does not switch to its new warming speed at once. It eases into it over the plant's measured bend time (about 20–30 min going into cooling or fuel switch, about 4 min into shutdown). We replayed the same 8 plant stretches as last time with six versions of the twin.



Average gap between twin and plant stack temperature over all 6,844 minutes of the 8 replays. Shorter is better. The grey bars are a version that failed: plant slopes everywhere drift in long cooling runs and overshoot after heat-up.

What actually helps

Letting the twin use the plant's own warming speed in the four warming stages (heat-up air, heat-up N_2+H_2, fuel switch, part load), keeping physics for standby and cooling, and stopping heat-up at the air-heater target cuts the gap from **56 °C to 24 °C**. Heat-up N_2+H_2 goes from 136 °C off to 4 °C off.

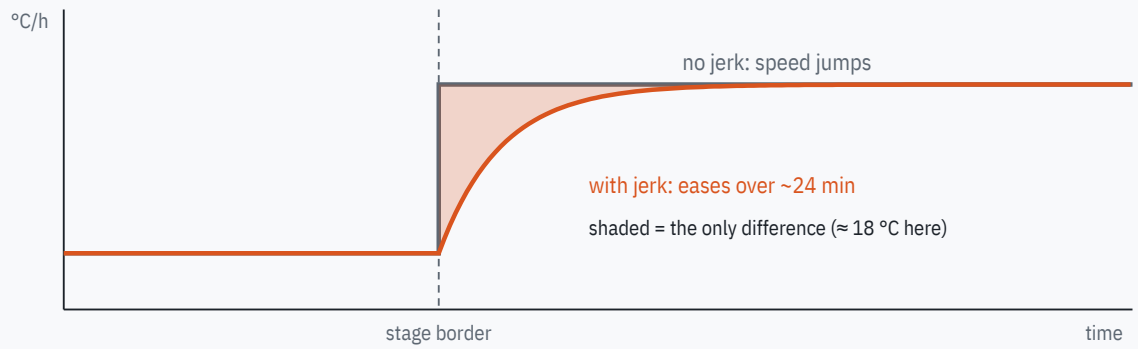
Page 2 shows what the jerk does at a corner and why it doesn't help here. Page 3 has every stage and the next decisions.

IF YOU REMEMBER ONE THING

> The jerk only shapes the first 10–30 minutes after a stage change. The twin's errors build up over hours because heat is missing, so the jerk cannot fix them.

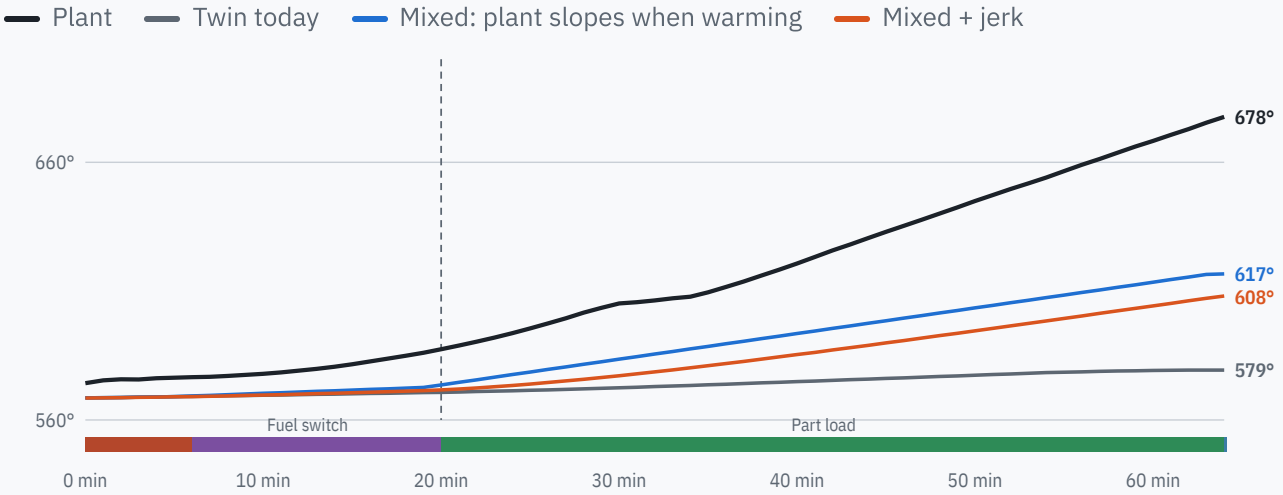
What the jerk does at a stage border

The jerk only changes how the warming speed turns the corner. Over a 24-minute bend into part load, that is worth at most about 18 °C, once.



Fuel switch → part load on the plant: the warming speed goes from about +18 to +115 °C/h and takes about 24 minutes to get there (measured on all 39 days). The shaded area is everything the jerk adds.

Fuel switch → part load, 25 March



Dashed line = start of part load. The plant bends up hard. The mixed twin bends too, but less. Adding the jerk delays the bend further, so the jerk version (orange) ends further below the plant: 608 °C against 617 without jerk and 678 measured.

Why it adds nothing on top of plant slopes

The plant slopes we give the twin are averages over each whole stage. Those averages **already include** the slow start of every stage. Adding the jerk on top counts the slow start twice, so the twin lags behind more.

The jerk would help only with “settled” slopes: the speed the plant reaches *after* the bend, not the stage average. That is a possible next test (page 3).

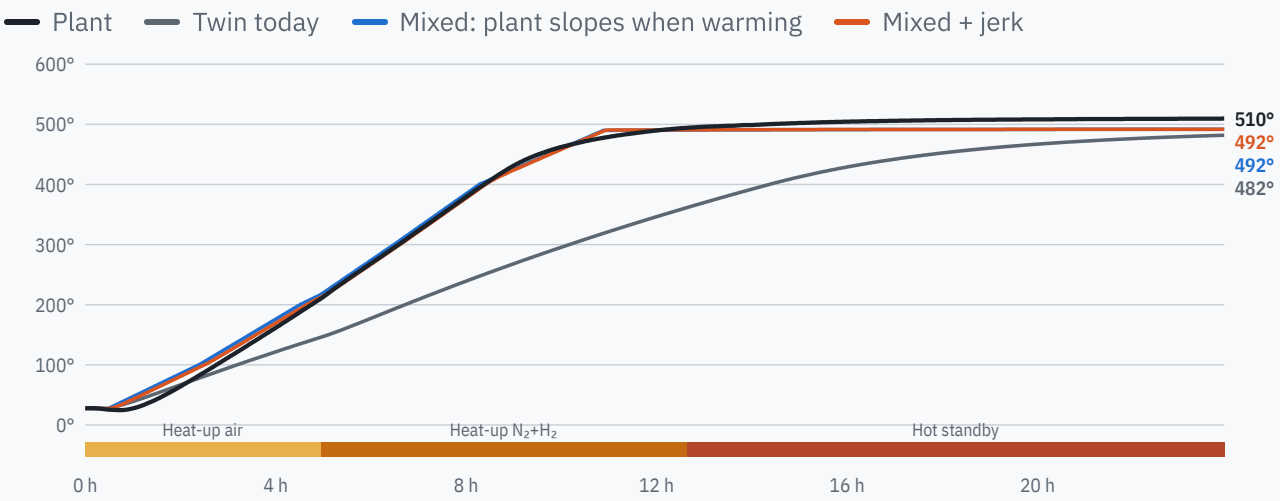
IF YOU REMEMBER ONE THING

The jerk adds a slow start to every stage. Our plant slopes already contain that slow start, so adding it again makes the twin lag.

Stage by stage, and what next

The jerk only helps in heat-up air: 14 → 9 °C, where the air heater starts from cold. Everywhere else it is neutral or slightly worse.

Cold start, 1 March



The mixed twin (blue, orange) follows the plant through the whole start-up and stops at hot standby. The two lines lie almost on top of each other, so the jerk barely shows.

Average gap to the plant, °C, per stage

Plant stage	Twin today	+ jerk	Mixed	Mixed + jerk
Heat-up air	21	22	14	**9**
Heat-up N_2+H_2	136	139	**4**	5
Hot standby	62	62	13	**13**
Fuel switch	10	10	**6**	8
Part load	44	45	**24**	32
Air-flow cooling	**43**	44	**43**	44
Shutdown	**7**	**7**	**7**	**7**
All minutes	**56.4**	**57.4**	**23.5**	**23.8**

Green = best in that row. Plant slopes and bend times were measured on the 39 days **without** these 8 replay stretches, so the test is fair. Exception: the 24-min fuel switch → part load bend comes from the earlier all-days study.

Still open

Cooling with 9000 ln/min air stays 81 °C off in every version. This needs the physics fix from the last report (stronger air heat exchange), not slopes or jerk.

Part load is still 24 °C off: the plant warmed about 120 °C/h on 25 March while the average for its band is 60–82.

Decisions for you

A. Put the mixed version into the 2D twin (plant slopes when warming, physics in standby and cooling)? It cuts the gap from 56 to 24 °C.

B. Keep the jerk out of the twin's temperature, and use it only to make the curves look right in the synthetic-data generator and for rate-of-change checks?

C. Test “settled slope + jerk” (the speed after the bend, plus the bend) as the cleaner way to use the jerk?

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Scripts: session scratchpad cmp/ — jerk_tables.py (slopes + bend times, replay days
left out), replay2.js (6 twin versions), score2.py, gen_jerk.py
```

> IF YOU REMEMBER ONE THING
>
> Use plant slopes in the warming stages, not the jerk. Keep the jerk for curve shape in the generator.